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# HOW PREDICTIVE AI GOES WRONG

IN 2015, administrators at Mount St. Mary's University, a private university in Maryland, USA, wanted to increase their university's retention rate—the proportion of admitted students who go on to graduate. The school conducted a survey to find which students were struggling. This might sound like a worthy goal; once the administration knows who is struggling, they can offer additional support to those students to help them adjust to college. But instead, the president suggested dismissing students who weren't doing well. He reasoned that if these students dropped out in the first few weeks of the semester, as opposed to later, they wouldn't count as “enrolled” and therefore wouldn't decrease the university's retention rate.

In a faculty meeting, the president said: “My short-term goal is to have 20–25 people leave by the 25th [of September]. This one thing will boost our retention 4–5%.”<sup>1</sup> Professors raised objections by pointing out that it is hard to judge who will be successful in the first few weeks after entering college. “This is hard for you because you think of the students as cuddly

bunnies, but you can't," the president responded. "You just have to drown the bunnies . . . put a Glock to their heads."

This is a startling example, but the fact is that many schools would like to predict which students are at risk of dropping out—some of them for reasons more aligned with students' well-being. An AI-based product called EAB Navigate could automate this process. In its marketing pitch to schools, EAB claimed: "The model will provide your school and its advisors with invaluable and otherwise unobtainable insight into your students' likelihood of academic success." Even if some schools might use this insight to pressure students to leave, others could conceivably use it to design interventions that might help students stay in school. But interventions that seem helpful could also be questionable. For example, the tool helps by recommending alternative majors in which a student would be more likely to succeed. This might have the effect of driving out poorer and Black students—whom the tool is more likely to flag—from more lucrative but more challenging STEM majors.<sup>2</sup> And throughout this process, students may have no idea that they are being evaluated using AI.

EAB Navigate is an example of an automated decision-making system that uses predictive AI. In this realm, there is a vast amount of AI snake oil.

Companies selling these tools make strong claims about their utility.<sup>3</sup> They are deployed widely by governments and private companies. Still, unlike generative AI applications like ChatGPT, predictive AI has largely flown under the radar when it comes to public interrogation. In many cases, including EAB Navigate, the decision subjects don't even know they are being automatically evaluated.

In this chapter, we will see how predictive AI goes wrong. While a full account of these failures would take up more than

the entire length of this book,<sup>4,5\*</sup> we will highlight common failures that are nevertheless hard to fix, underscoring the difficulty of making predictive AI work.

Before we dive in, let's look more closely at how these automated decisions are made.

## Predictive AI Makes Life-Altering Decisions

Algorithms like EAB Navigate are everywhere, and they are used to automate consequential decisions about you—with or without your knowledge.<sup>6</sup> When you show up at a hospital, an algorithm could determine if you need to be admitted overnight or released the same day. When you apply for public benefits such as child welfare, an algorithm could determine if your application is valid or if you are trying to commit fraud. When you apply for a job, an algorithm could determine if your application will be considered by a recruiter or discarded. When you visit a beach, an algorithm could determine if the water is safe to swim in.

An algorithm is a set of steps or rules used to make a decision. Sometimes these rules are developed by people or institutions. For example, during the COVID-19 pandemic, the U.S. government sent stimulus checks to help citizens deal with the economic hardships they faced. Adults were paid USD 1,200; children were paid USD 500. Policymakers decided these rules. But once the rules were established, people's eligibility was automatically determined based on their past tax records.

\* Two popular incident databases record failures of AI, including predictive AI, in the real world. The AI Incident Database has over 600 reports, and the AI, Algorithmic, and Automation Incidents and Controversies Repository has over 1,400 reports, as of 2024.

1. If the applicant is a U.S. citizen and eighteen or older, send a check for USD 1,200.
2. If they are a U.S. citizen and under eighteen: send a check for USD 500.
3. If they're not a U.S. citizen or earn more than USD 75,000 per year\*: don't send a check.

This type of algorithm, in which the rules are developed manually but applied automatically, is commonly employed in the public sector, such as in allocating funds for public housing or welfare.

But increasingly, algorithms are used to *develop* the rules automatically from patterns in past data. For instance, while watching Netflix, if you rated *Forrest Gump* highly but not *The Shining*, a recommendation algorithm might predict that you'll enjoy drama films over horror films. Here, a Netflix employee is not manually creating a rule stating that users who liked *Forrest Gump* should be recommended other drama films. Instead, your ratings and the time you spend watching different types of films determine what films you're recommended next. Unlike the first kind of algorithm, the rules are both developed and applied automatically.

This kind of algorithm is called a model, which is a term you might encounter frequently. A model is usually a set of numbers that mathematically specify how the system should behave. These numbers may not be intelligible to a human, even to the developer of the system, unless the model is explicitly designed with that goal in mind. Models are created from data, or "trained," using a set of statistical techniques called machine learning.

\* The actual algorithm was slightly more complex: the amount paid if a person earned more than USD 75,000 tapered off. And there were different rules for people with children.

These models can be much more consequential than deciding your next Friday night movie. They are used for allocating scarce resources, such as jobs or loans. They provide certain opportunities to people and foreclose other opportunities. This is what we call predictive AI: models used for decision-making based on predictions about the future, such as who will do well at a job or who will pay back a loan.

Take the example of criminal justice, where predictive AI is used for many kinds of decisions: Should an inmate get parole? What should happen to people who are arrested? Before trial, a judge needs to decide whether a defendant should be detained in jail, released on bail (and if so, what the bail amount should be), or released without any bail, but perhaps with other restrictions such as ankle monitors.

A lot rides on the answer.<sup>7</sup> Spending time in jail can ruin someone's life. They can lose their income in the short term—and even if they are released, especially if they have restrictions such as ankle monitors, it will still be much harder to find a job. They suffer from increased physical and mental illnesses, both due to stigmatization and because of poor conditions in jail. And a large fraction of the people in jail are simply there because they can't afford high bail amounts.

As a result, the criminal justice system disproportionately burdens the poor and leads to cycles of poverty and racial inequality. Almost half a million people are in U.S. jails at any given time without having been convicted of a crime.<sup>8</sup> Despite violent crime in the country going down by almost 50 percent, the number of people jailed in the last four decades has nearly doubled.<sup>9,10</sup>

Many U.S. states mandate the use of risk assessment tools to decide whether a person should be released or detained before trial, and predictive AI is often used. The tools produce two

main risk scores. One is the risk that the defendant will commit a crime, especially a violent crime, if released. The other is the risk that they will fail to appear in court on the appointed date. In each case, defendants are labeled low, medium, or high risk. The tools try to calculate these scores based on certain features of the defendants.

Let's zoom in on one predictive AI product: COMPAS, which stands for Correctional Offender Management Profiling for Alternative Sanctions. COMPAS relies on defendants' answers to 137 questions to make decisions about them.<sup>11</sup> It includes questions about their past history of crime or failure to appear for court cases. It also includes questions about things individuals have little or no control over, such as how often their family members have been arrested, or whether their friends or acquaintances have been arrested. And some questions seem to punish people for their personality or for poverty, such as "How often do you feel bored?" and "How often do you have barely enough money to get by?"

The developers of COMPAS aim to predict if a defendant will fail to appear in court or be arrested for a crime within two years.<sup>12</sup> It is trained using data on defendants' behavior in the past. COMPAS uses this past data to find patterns in the characteristics of defendants who failed to appear in court, such as their age, number of past offenses, and the criminal history of their peers, and tries to distinguish them from those who did appear for their trial. This highlights an assumption built into much of predictive AI: people with similar characteristics will behave similarly in the future.

Predictive AI is quickly gaining in popularity. Hospitals, employers, insurance providers, and many other types of organizations use it. A major selling point is that it allows them to reuse existing datasets that have already been collected for

other purposes, such as for bureaucratic reasons and record keeping, to make automated decisions.

Yet, predicting the future is hard. People could face unexpected setbacks such as getting evicted, or surprises such as winning a lottery, that no model can predict. Small changes in someone's life, such as a visit to the emergency room, could have large effects on their future—say, due to compounding medical bills.

Claims about the virtues of predictive AI are pervasive. Upstart's model decides whether an applicant should be approved for a loan.<sup>13</sup> The company claims that its model is significantly more accurate than traditional lending models. It also claims to be a leader in fair lending practices and promises that future versions of the model will continue to be fair. Finally, it claims to be highly efficient: three-quarters of its loan decisions require no input from humans. Other companies make similar promises. HireVue sells tools to automate hiring decisions. It claims to predict how well a candidate will perform at their job once hired. HireVue's home page announces its selling points: "Fast. Fair. Flexible. Finally, hiring technology that works how you want it to."

Despite these statements, human decisions still come in at many stages of the development of predictive AI, often hidden from sight. And since the data used to train predictive AI is itself created using human discretion, we cannot guarantee that the decisions will be unbiased or fair.<sup>14</sup> In other words, decisions made using predictive AI may still be very human.

We were skeptical of the promises made by developers of predictive AI, and so we decided to dig deeper. Together with researchers Angelina Wang and Solon Barocas, we spent over a year reading hundreds of research papers, news articles, and

reports about automated decision-making systems. We were surprised to find that the same set of shortcomings plague many applications of predictive AI.<sup>15</sup> In the next few sections, we will take a closer look at these shortcomings, through real-world case studies.

Spoiler: We think predictive AI falls far short of the claims made by its developers.

### A Good Prediction Is Not a Good Decision

Healthcare workers make a critical decision when a patient comes to the hospital with symptoms of pneumonia: whether to send the patient home after treatment or admit them overnight. They consider a patient's age as well as preexisting medical conditions, such as asthma, which place them at high risk if they get pneumonia. High-risk patients with pneumonia symptoms are typically sent straight to the ICU to avoid complications.<sup>16</sup>

In a 1997 study, researchers investigated whether AI could make better decisions than healthcare workers in predicting outcomes of patients with pneumonia.<sup>17</sup> Like countless other AI researchers, they thought a model trained with a sufficiently large amount of data would make better decisions than a human, decisions which would help to prioritize high-risk patients.

The researchers trained an AI model and found that it had fairly good accuracy in predicting which pneumonia patients would face complications or death. Surprisingly, the model found that having asthma leads to *lower* risk of complications due to pneumonia. If this model were to be used in a hospital, it would be more likely to send an asthmatic patient home than a non-asthmatic one (let alone sending asthmatic patients to the ICU). How is this possible?

The researchers looked more closely. They found that in the dataset, asthmatic patients were indeed at a lower risk for serious pneumonia or death—but this was only because the training data was collected under the hospital’s preexisting decision-making system.<sup>15</sup> Asthmatic patients were sent straight to the ICU as soon as they arrived, thereby receiving more intensive care than non-asthmatic patients and thus becoming less likely to develop complications.

So the model’s predictions were correct—but they were correct under the hospital’s existing system. Ironically, the model was meant to replace this system.

Deploying this model would have led to disastrous decisions. Asthmatic patients would have been perceived as lower risk by the model and sent home. Thankfully, the researchers realized what had happened and did not use the model in this particular hospital.

This error shines a light on a fundamental limitation of much of predictive AI: AI can make good predictions *if nothing else changes*. But correlation is not causation (and having asthma does not reduce the risk of complications from pneumonia). In other words, predictive AI does not account for the impact of its own decisions. That is, it can’t tell us what would happen if something changed in a system—if, for instance, the model started sending asthma patients home.

Let’s look at another example from healthcare. A 2018 study claimed to accurately predict hypertension using machine learning, with impressive results.<sup>18</sup> On closer look, however, it became clear that the model was evaluated on people who had already been seen by clinicians.<sup>19</sup> This meant that a critical input to the model was whether a patient already used drugs for controlling hypertension. But if a patient is using antihypertensive drugs, they have obviously already been diagnosed with hyper-

tension. These cases were still counted as successes when evaluating the model, vastly inflating its accuracy numbers.

This issue arose in part because the researchers relied on existing data rather than new data collected specifically for the task. Generally, collecting data is expensive and time-consuming. So, some AI developers claim that existing data is enough to make good decisions.

The importance of collecting new data is widely recognized in medicine, which relies on randomized controlled trials (RCTs) to test the impact of introducing a new drug or vaccine. To test if a drug performs well, half the people in the trial are randomly assigned to the experimental group and are given the drug. The other half are assigned to the “control” group and are given a placebo. The groups are carefully balanced with respect to demographics like age and gender. A drug’s effectiveness is evaluated based on whether it leads to lower disease rates in the experimental group compared to the control. Medical researchers perform RCTs despite their slow pace and high expense for a simple reason—easier, faster methods don’t work. The same is true in many areas where AI is used for automated decision-making.

Unfortunately, predictive AI companies don’t seem to have realized the importance of collecting data on the impact of their tools’ decisions. And such data collection is also expensive and time-consuming, so it would negate their claims of cutting costs and increasing efficiency.

As a result, even if AI can make accurate *predictions* based on past data, we can’t know how good the resulting *decisions* will be before AI is deployed on a new dataset or in a new setting. So, when you encounter claims about the effectiveness of predictive AI, it’s important to find out if developers evaluated the impact of the decisions or only the accuracy on past data.

## Opaque AI Incentivizes Gaming

We've seen that we can't fully determine the impact of AI before deployment, in part because AI makes its predictions assuming that the system will remain the same as it was during testing. But systems change, and people are one obvious wild card. Things become even more complicated when people behave strategically.

The colonial British government in India wanted to reduce the cobra population, so it decided to offer a reward to people who brought in dead cobras. But instead of killing cobras in the wild, people started breeding them to claim the reward—leading to an *increase* in the cobra population. This is an apocryphal example of a mismatch between what is specified and what is desirable. A similar problem occurs in AI. When developers build AI, they specify the exact outcome they want to predict. But because of the reliance on past data, they are forced to choose this outcome based on the data that is already available. As a result, what AI predicts can be a poor proxy for what we want it to predict.

A stark example is AI used for hiring. In the United States, three-fourths of employers use automated tools to screen job candidates.<sup>20</sup> Automated hiring tools come in various forms. Some are used to filter candidates based on their résumés. Others test candidates using an automated video interview. Yet others require them to solve puzzles. These tools are used as the first step in selecting candidates. If someone doesn't pass, their résumé is discarded without a human ever looking at it. But the process is opaque; companies don't share how their software is built, and candidates are left in the dark about what criteria they are judged on.<sup>21</sup>

In response, candidates have developed strategies to work around opaque hiring AI. They stuff their résumés with keywords

from the job application and add the names of top universities in white text (which a human reader can't see, but a computer can recognize).<sup>22</sup> In video interviews that they know will be judged by AI, they try using fancy words such as "conglomerate."<sup>23</sup>

It is not clear whether these strategies work. So a group of journalists decided to investigate. They looked at Retorio, a Munich-based startup that offers hiring tools based on video interviews.<sup>24</sup> Their findings were surprising: simple changes to a person's appearance, such as wearing a scarf or glasses, were enough to drastically change the score that the AI tool produced. Adding a bookshelf or a painting in the background increased scores, while making a video darker (without changing its content) led to lower scores. In another study, researchers looked at personality tests used in hiring and found that merely changing the format of candidates' résumé from PDF to plain text changed their personality scores.<sup>25</sup>

Changing the background or format of a résumé obviously does not change someone's capability to perform well at a job. So why do candidates' scores change? One reason could be that in the data used to train the model, people who had bookshelves in the background performed better compared to those who had a plain background.

This is precisely the kind of correlation that candidates banked on when they added keywords to their résumé. They assumed people with those keywords in their résumé performed better in the past. They wanted to avoid getting rejected just for omitting the right keywords or not using enough fancy words.

Acting strategically—gaming—to avoid negative consequences is common. This is what teachers do when they teach to the test, and what consumers do when they try to increase

their credit scores without changing their spending habits, such as by getting a retail credit card or filling out a prequalification form before applying for credit. But with AI-based hiring, people don't even know what actions influence their chances. Instead of guiding candidates toward decisions that actually improve their skills, AI-based hiring encourages candidates to experiment with superficial changes to their résumé and application.

We are not taking a stance on whether candidates should or shouldn't be attempting these tricks. The ethics of gaming are interesting but tangential to our point. Rather, our point is that when AI companies make claims about accuracy, they fail to account for the effects of people behaving strategically. When the model's output can be easily manipulated using superficial changes, we cannot take claims of accuracy at face value. Opaque models also impose a cost on decision subjects in terms of wasted time, such as when candidates experiment with ways to add invisible fake qualifications to their résumés.

### Overautomation

In 2013, the Netherlands deployed an algorithm to flag welfare fraud, replacing an earlier system in which humans reviewed each decision.<sup>26</sup> The algorithm was used to make serious accusations of guilt using only statistical correlations in data, without any other evidence.<sup>27</sup>

This shift to an automated system had many adverse effects. For one, people lost the ability to challenge decisions. Inaccurate or outdated government data often leads to erroneous fraud allegations, which under the new system could not be reversed. Further, the data used to make these allegations was not publicly available, so people had no way to find out why they'd been accused of fraud.

Over the next few years, the algorithm wrongly accused around thirty thousand parents of welfare fraud in childcare payouts. The amounts the government said they owed were in some cases over a hundred thousand euros—sending many parents into mental and financial ruin.<sup>28</sup> Shockingly, the algorithm used nationality to predict if someone had committed fraud. All else being equal, people were more likely to be flagged for fraud if they had Turkish, Moroccan, or Eastern European nationality.<sup>29</sup>

Despite the algorithm's shortcomings, the country used it for six years. When details of the algorithm were shared in 2019, the public was outraged. The Dutch data protection watchdog investigated privacy failures caused by the algorithm's use. It fined the country's tax authorities—who had built the algorithm—EUR 3.7 million. This was the largest such fine in the country. In 2021, the prime minister and his entire cabinet resigned, in part due to the use of the welfare fraud algorithm.

This is an example of overautomation: when AI is used to make decisions without offering any recourse to decision subjects. Overautomation has led to other notable failures in fraud detection—even when AI is not in use. Using an algorithm to detect unemployment fraud, the U.S. state of Michigan wrongly collected USD 21 million from residents between 2013 and 2015.<sup>30</sup> And the Australian government incorrectly collected AUD 721 million from its citizens between 2016 and 2020 in what's been called the Robodebt scandal.<sup>31</sup>

To protect themselves from accusations of overautomation, developers of these systems often include fine print saying that they should always be used with human supervision. But this merely passes the buck and doesn't necessarily have the intended effect.

In the summer of 2022, Toronto used AI to predict when people should avoid swimming at public beaches due to high bacteria levels.<sup>32</sup> The developer claimed the software achieved over 90 percent accuracy in predicting water safety. But it did much worse: on 64 percent of the days when the water was unsafe, beaches remained open based on incorrect assessments.

When journalists questioned city officials about the efficacy of the predictive tool, officials responded by saying it wasn't used on its own—a human overseer always made the final decision. Journalists later found that the supposed overseers *never* changed the decisions that the software spit out.

This is a familiar pattern. AI developers use a bait and switch when it comes to human oversight. They sell predictive AI based on the promise of full automation. Eliminating jobs and saving money is a big part of their pitch. But when AI fails, developers retreat to the fine print, saying that it shouldn't be used without human oversight.

Even if that oversight technically exists, it is often inadequate due to limitations of time, expertise, or authority. The bureaucrats in charge may be overworked, or may not have the training to challenge the automated decisions or the incentive to stick their necks out to do so.

In one extreme case, U.S. health insurance company UnitedHealth forced employees to agree with AI decisions even when the decisions were incorrect, under the threat of being fired if they disagreed with the AI too many times. It was later found that over 90 percent of the decisions made by AI were incorrect.<sup>33</sup>

Even without such organizational failure, overreliance on automated decisions (also known as “automation bias”) is pervasive. It affects people across industries, from airplane pilots to doctors. In a simulation, when airline pilots received an

incorrect engine failure warning from an automated system, 75 percent of them followed the advice and shut down the wrong engine. In contrast, only 25 percent of pilots using a paper checklist made the same mistake. If pilots can do this when their own lives are at stake, so can bureaucrats.<sup>34</sup>

No matter the cause, the end result is the same: consequential decisions about people's lives are made using AI, and there is little or no recourse for flawed decisions.

### Predictions about the Wrong People

AI reflects its training data. It learns patterns about the people who make up the data, and the decisions made by AI reflect these patterns. But when the decision subjects come from a population with different characteristics than those in the training data, the model's decisions are likely to be wrong. For instance, a predictive AI system that performs well in one country could be utterly useless in another.

Let's look at how this plays out in two criminal risk prediction systems in the United States: the Ohio Risk Assessment System (ORAS) and the Public Safety Assessment (PSA). Similar to COMPAS, both attempt to predict the risk of releasing a defendant before trial.

ORAS was trained on data from 452 defendants from the U.S. state of Ohio in 2010 and is now used across the United States. There are a few obvious problems with this. Ohio's patterns of criminal activity might differ from those in other states. The data doesn't represent everyone. The small number of people who were used to create the model could otherwise vary from the larger population it was eventually used on. Finally, as with any model, accuracy could degrade as patterns of crime change over time.<sup>35</sup>

By contrast, PSA was trained using 1.5 million people from three hundred jurisdictions across the United States. It is used in over twenty states. At first glance, this seems to solve part of the problem. If the model is trained on a much larger dataset from many jurisdictions, it should be more accurate when applied to the entire country. But nationwide trends of criminal activity can differ from local trends.

Cook County in Illinois is a stark example. It adopted PSA in 2015. But the rate of violent crimes there is much less than the national average. Compared to the training data, *ten times* fewer defendants who were marked as “high risk” went on to commit violent crimes. PSA used data from all over the country, but it did not consider that crime rates could be much lower in some counties. Thousands of defendants were unnecessarily jailed for months before trial—based only on a model’s prediction, without any evidence of their guilt.<sup>36</sup>

PSA erred because it didn’t distinguish data from different counties—it made predictions about the wrong people. In some cases, getting data about the entire population of interest is impossible, making this error hard to resolve.

One example comes from Allegheny County, Pennsylvania. In 2016, the county adopted the Allegheny Family Screening Tool to predict which children were at risk of maltreatment.<sup>37</sup> The tool is used to decide which families should be investigated by social workers. Through these investigations, social workers can forcibly remove children from their families and place them in foster care.

The tool relies on public welfare data, which consists mainly of data on poorer parents who use public services such as Medicaid-funded clinics. Notably, it doesn’t include information about people who use private insurance.<sup>38</sup> Models built using this data therefore cannot make decisions about rich par-

ents who have never relied on public services. As a result, the tool disproportionately targets poorer families.

This is an example of how AI tools search under the streetlight. More often than not, the streetlight is pointed at the poor.

Whenever predictive AI is deployed, it is critical to ask: Who was it tested on? When predictive AI is built using one population but is used on another, claims about how well it performs are based on insufficient evidence.

### Predictive AI Exacerbates Existing Inequalities

The cost of flawed AI is not borne equally by all. The use of predictive AI disproportionately harms groups that have been systematically excluded and disadvantaged in the past.

A notable example comes from predicting who should receive better medical care. With the passage of the Affordable Care Act in the United States in 2010, insurance companies started asking hospitals to provide services at lower prices, threatening to remove providers who did not comply. One of the main ways hospitals cut costs was by identifying high-risk patients and providing them with preemptive care so as to avoid future expenses, such as hospitalization. To identify high-risk patients, they turned to AI.

Dozens of models were developed to assess patients' health risks. Developers claimed that AI could rank patients based on their healthcare needs and assign additional resources to the ones classified as high risk. While the pneumonia risk prediction model that we saw in the beginning of this chapter was never deployed, health risk prediction tools used to reduce hospital expenses are deployed across the United States.

Enrollment in a high-risk healthcare program has huge consequences. It dictates whether a patient will receive preemptive

care and personal assistance. Still, most developers did not share how they built their models—until recently, we knew little about how well their products performed.

One such product is Optum's Impact Pro. In 2019, researchers studied it and found that based on its predictions, Black people were less likely to be admitted to high-risk programs than White people. In other words, a Black person who had the same health risks as a White person would receive poorer care.<sup>39</sup>

On closer inspection, the researchers found the culprit. It is hard to quantify how much a patient needs healthcare. But hospitals know exactly how much people *spend* on healthcare. So instead of predicting healthcare *needs*, Optum chose to predict how much an insurance company will *spend* on healthcare.

Higher healthcare costs don't always reflect higher health needs. Higher bills could occur due to access to better insurance; more time, attention, and care at the doctor's office; or more visits to the doctor. Or perhaps those with higher bills are simply those who can afford to pay higher costs for co-pays and deductibles. There are clear inequalities in the U.S. healthcare system, and the use of Optum's predictive AI would mean that people who were already receiving better healthcare would be more likely to be classified as high risk and continue to receive better care in the future.

Unsurprisingly, in the training data used for creating the model, Black people received poorer care than White people, even those with similar health conditions. The tool had a racial bias because the predicted quantity (healthcare costs) was a poor stand-in for what the developers claimed to predict (healthcare needs or a patient's risk level).

Optum's use of healthcare costs to measure risk made sense for its business. Its clients were hospitals that wanted to reduce

costs. Even after this damning study was published, the company continued using healthcare costs to build its model.

Business incentives are one of many reasons predictive AI increases inequality. Another is developers' reliance on past data. As we have seen, collecting new data for developing predictive AI is expensive and time-consuming. But existing data might not have information about what developers want to predict (in this case, patients' healthcare needs or their risk levels). So, developers use proxies that are easier to measure and already exist in the data (such as healthcare costs).

Let's look back at COMPAS, the tool which provides a pre-trial risk score. Its developers claim that judges can use this score to predict whether a defendant will commit a crime or fail to show up for their trial. But the data used by COMPAS does not contain information about crimes. It only includes information on who was arrested. The difference is subtle but salient. Not all crimes lead to arrests—in many cases, they can be undetected or ignored. Police could arrest someone only to later find them innocent in court. And there are well-known racial disparities in policing—Black people in the United States are more likely to be arrested than White people for the same crimes. These differences can create chasms between what companies claim to predict (crime) and what they actually predict (arrests).

For all these reasons, when predictive AI systems are deployed, the first people they harm are often minorities and those already in poverty. We've already seen other examples, such as Allegheny County's child maltreatment prediction tool that only had data about those who access public benefits such as child welfare. And we'll see many more in the chapters to come.

## A World without Prediction

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Why is predictive logic so pervasive in our world? We think a major reason is our deep discomfort with randomness. Many experiments in psychology show that we see patterns where none exist, and we even think we have control over things that are, in fact, random.<sup>40,41</sup> When people are forced to confront randomness and the illusion of control breaks down, they look for ways to take back control.

A good example is election forecasting. In the United States, presidential elections occur every four years, and forecasting the winner is a common practice, starting well over a year in advance. But it is far from an exact science. On the morning after the 1948 presidential election, the *Chicago Tribune* infamously blared “Dewey Defeats Truman” on its front page.<sup>\*42</sup> Of course, Truman had won the election and would be the next U.S. president. In a hurry to call the election before the paper went to press, the newspaper relied on poll outcomes to preemptively predict the (incorrect) winning candidate. And seventy-five years later, things are only slightly better. The 2016 election forecasts were famously incorrect, predicting Hillary Clinton defeating Donald Trump.

Despite its limitations, forecasting has become a sort of spectator sport, and people’s fascination with it can border on obsession. Here’s a quote from a *Wired* article about one such person:

*When he wakes up in the morning, he doesn’t shower or eat breakfast before checking the Nate Silver-founded site’s presi-*

\* Harry S. Truman and Thomas E. Dewey were the Democratic and Republican presidential candidates in the 1948 elections.

*dential election forecast. He keeps a tab open to FiveThirtyEight's latest poll list; a new poll means new odds in the forecast. He gets push alerts on his phone when the forecast changes. He follows the 538 Forecast Bot, a Twitter account that tweets every time the forecast changes. In all, Evan says he checks in hourly, at least while he's awake.<sup>43</sup>*

This is despite poll numbers not affecting most people's daily lives in any meaningful way. Why does this happen? It all comes down to our inability to handle the uncertainty hanging over our lives. Uncertainty avoidance goes one level deeper than being averse to risk. Even if the election forecaster says the chance of our favorite candidate winning is 50-50, that somehow feels better than not knowing at all.

What does this have to do with AI snake oil? Here's our thesis. It's true that companies and governments have many misguided commercial or bureaucratic reasons for deploying faulty predictive AI. But part of the reason surely is that decision-makers are people—people who dread randomness like everyone else. This means they can't stand the thought of the alternative to this way of decision-making—that is, acknowledging that the future cannot be predicted. They would have to accept that they have no control over, say, picking good job performers, and that it's not possible to do better than a process that is mostly random.

And what if a decision-maker did embrace unpredictability? Suppose a company announced that it would hire candidates randomly after filtering out clearly unqualified candidates, and it would promote employees randomly among those who met performance criteria. Because the overemphasis on merit is so ingrained in so many people, the company would come to be seen as an undesirable place to work, and it would lose

good candidates. Such a policy wouldn't survive long in our world.

In fact, when such lotteries were used for allocating housing, people participating in those lotteries had a largely unfavorable opinion of them.<sup>44</sup>

And so, the discomfort we experience with randomness can lead to a search for patterns where none exist. This paves the way for biases. Consider the business mantra, "No one ever got fired for buying IBM." Opting for the familiar, the tried-and-true, is safer than embracing uncertainty. Even the practice of hiring primarily from elite colleges reflects a desire to impose order on the inherently unpredictable process of identifying talent and potential.

Accepting the inherent randomness and uncertainty in many of these outcomes could lead to better decisions, and ultimately, better institutions. Instead of treating people as fixed and their outcomes as predetermined, we need to work on building institutions that are genuinely open to the fact that the past doesn't predict the future. Such a world is possible—if only we can learn to embrace the randomness that underpins our lives. We'll return to this theme in the final chapter.

## Concluding Thoughts

As predictive AI is deployed, it is important to understand how it fails and how it harms people, in order to avoid falling prey to snake oil. Change starts by challenging the deployment of harmful AI tools in your workplace, neighborhood, and community. An informed public is the first step in demanding change.

In this chapter we encountered many reasons why predictive AI fails. Table 2.1 recaps them. But given the increasing amount

TABLE 2.1. Five reasons predictive AI fails

| Reason                                                                                | Example                                                                                                                  |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| A good prediction can result in a bad decision.                                       | Patients with asthma could be sent back home when they come to a hospital with symptoms of pneumonia.                    |
| People can strategically game opaque AI.                                              | Adding bookshelves in the background increases scores on automated hiring tools.                                         |
| Users over-rely on AI without adequate oversight or recourse.                         | The Dutch welfare fraud detection model falsely accused 30,000 parents of fraud without any recourse.                    |
| Data for training AI may come from a different population than the one it is used on. | PSA's criminal risk prediction relied on a national sample. It overestimated the risk in counties where crime was rarer. |
| Predictive AI can increase inequality.                                                | Optum's Impact Pro led to an increase in the difference in the quality of care between Black and White patients.         |

of data collected about people, as well as advances in machine learning, it might seem like the limitations we have seen are temporary. On the other hand, it is also possible that no matter how much data we have or how good our models become, there are inherent limits to how predictable the future is. Which of these scenarios is more likely? The next chapter will answer this question.